ECOCYCLE – E-WASTE COLLECTION & RECYCLING MANAGEMENT

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

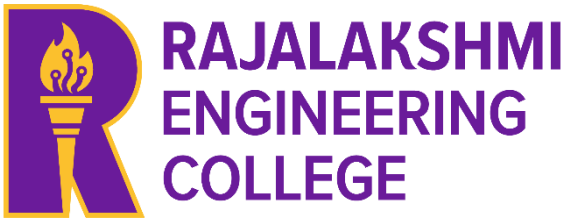
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE ENGINEERING



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BONAFIDE CERTIFICATE

Certified that this project report titled "EcoCycle – E-Waste Collection & Recycling Management" is the bonafide work of Jenell S G (Roll No. 240701212), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, Jenell S G (Roll No. 240701212), hereby declare that the project report titled "EcoCycle – E-Waste Collection & Recycling Management", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

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JENELL S G

ABSTRACT

Electronic waste such as computers, printers, monitors and laptops must be collected and recycled, and items that are old, damaged or hazardous need attention first. Without a single record of what the waste is, who has asked for it to be collected and what happened afterwards, urgent items are missed and recycling cannot be followed.

EcoCycle is an e-waste collection and recycling management application built on Zoho Creator. Its E-Waste Entry, Collection Request and Recycling Tracking forms capture the waste, the collection requests and the recycling activity. A Deluge form workflow calculates a priority score from quantity, age, hazardous material and condition and classifies each entry as High (8 or above), Medium (4 to 7) or Low (0 to 3); new entries start with collection status Pending and recycling status Not Processed, collection requests with past dates are prevented, and an e-mail notification is sent for high-priority e-waste. Four reports and the EcoCycle Dashboard, with a priority pie chart and two status bar charts, show the position. Each collection request creates a lead in Zoho CRM, a read-only Sentiment field labels the remarks of a request, a Custom API, EcoCycle Summary API, is secured with OAuth2, and a Zia chat agent, the EcoCycle Data Assistant, answers questions about the data with read-only access.

The report is limited to what the project evidence shows. The forms, reports, dashboard, CRM leads and one chat exchange are shown in screenshots, and the levels on the dashboard agree with the stated ranges. The priority weights, the Deluge sources and the e-mail output were not supplied, so the code given is a reference implementation, and the built-in Creator REST API integration is planned and not implemented.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; E-Waste Management; Priority Scoring; Zoho CRM Integration; Zia Chat Agent

PROJECT AT A GLANCE

Attribute	Details
Project Title	EcoCycle – E-Waste Collection & Recycling Management
Application Domain	Environment – e-waste collection and recycling management
Student Name & Register No.	Jenell S G – 240701212
Zoho Creator Edition	Trial edition (13 days remaining when the screenshots were taken)
Application Name (Link Name)	ecocycle (from the browser address)
Published / Live URL	https://creatorapp.zoho.in/jenell.sg.2024.cse_rajala718/ecocycle (address seen in the browser)
Demo Credentials (per role)	Owner account only (no separate test accounts recorded)
Number of Forms / Subforms	3 forms (E-Waste Entry, Collection Request, Recycling Tracking); 0 subforms seen
Number of Reports (by type)	4 reports (All E-Waste, High Priority E-Waste, Collection Requests, Recycling Tracking Report) and 3 dashboard charts (1 pie, 2 bar)
Number of Pages / Dashboards	1 dashboard page (EcoCycle Dashboard) and the AI Agents page
Number of Workflows	Form: 4 (Table 5.3) Schedule: 0 Approval: 0 Blueprint: 0 Batch: 0
Deluge Functions / Custom Actions	4 form-workflow scripts (reference implementations in Appendix C)
Integrations	Zoho CRM (connection and lead creation); Custom API EcoCycle Summary API (OAuth2); Zia chat agent (EcoCycle Data Assistant)
Roles & Profiles Defined	None defined (single owner account)
Environments Used	Single application instance (development, stage and production not used)
Date of Production Deployment	Not deployed to a separate production environment (trial account)

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CRM	Customer Relationship Management
DS	Deluge Script
ER	Entity–Relationship
FR	Functional Requirement
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
QR	Quick Response
REST	Representational State Transfer
SMS	Short Message Service
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator
WF	Workflow
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Computers, printers, monitors and laptops reach the end of their useful life in every institution. Electronic waste is not ordinary waste: it can contain hazardous material, it must be collected from the place where it is stored, and it should be sent to a recycler rather than discarded. The Global E-waste Monitor 2024 reports that 62 million tonnes of e-waste were generated worldwide in 2022 and that only 22.3 per cent of it was documented as formally collected and recycled [3]. Items that are old, damaged or hazardous need attention sooner than items that still work, but without a record there is no way to tell which is which.

EcoCycle is an e-waste collection and recycling management application built on the Zoho Creator low-code platform. It records each e-waste item, calculates a priority score and level from four factors, accepts collection requests, tracks recycling, e-mails an alert for high-priority items, creates a lead in Zoho CRM for each collection request, shows the position on a dashboard and offers a Zia chat agent, the EcoCycle Data Assistant, that answers questions about the stored data.

1.2 Problem Statement

E-waste handling needs three things to be kept together: what the waste is and how urgent it is, who has asked for it to be collected and when, and what happened to it after collection. The problems the project addresses are therefore:

- E-waste details such as quantity, age, hazardous content and condition must be recorded in one place.
- The urgency of each item must be decided in the same way every time, without manual judgement.
- Collection requests must be captured with the requester, location, preferred date and time, and their status followed.
- Recycling activity must be tracked from Not Processed onwards, with the recycling method and recycler recorded.
- Requests for collection should reach the organisation's customer-relationship records, and staff should be able to ask for information in plain language.

The project description does not describe an earlier process, so the report treats EcoCycle as a new system and makes no claim about how e-waste was handled before it.

1.3 Objectives

- Store e-waste entries with category, quantity, age, hazardous-material flag, condition and collection location.
- Calculate a priority score from quantity, age, hazardous content and condition, and classify it as High (8 or above), Medium (4 to 7) or Low (0 to 3).
- Initialise the collection status as Pending and the recycling status as Not Processed for every new entry.
- Reject collection requests whose preferred date is in the past.
- Send an e-mail notification for high-priority e-waste.
- Provide four reports: All E-Waste, High Priority E-Waste, Collection Requests and Recycling Tracking.
- Provide a dashboard with a priority pie chart, a collection-status bar chart and a recycling-status bar chart.
- Create a lead in Zoho CRM automatically when a collection request is submitted.
- Provide a read-only Zia chat agent that answers questions about the application data.

1.4 Scope of the Project

In scope: three data forms (E-Waste Entry, Collection Request and Recycling Tracking); form workflows for priority calculation, status initialisation, date validation and e-mail notification; four reports; the EcoCycle Dashboard page; a connection to Zoho CRM that creates a lead for each collection request; a Sentiment field on the Collection Request form; a Custom API, EcoCycle Summary API; and the EcoCycle Data Assistant chat agent with read-only access.

Out of scope or not yet built: the Zoho Creator REST API integration (planned as a future enhancement and not implemented), Google Maps integration, automated collection scheduling, SMS notifications, QR-code identification and real-time collection tracking. These are listed in Chapter 9. Roles and permission profiles were not described.

1.5 Justification for a Low-Code Approach

EcoCycle is made of forms, reports, a dashboard and a small number of rules, which are the building blocks Zoho Creator provides directly. The priority calculation, the status defaults, the date check and the e-mail alert are form workflows written in Deluge. The link to Zoho CRM is a connection configured under Microservices, and the chat agent is configured on the platform [4], [5], so no separate server or machine-learning service was needed.

The trade-offs are discussed in Chapter 8: the platform limits what can be changed, the application depends on the platform's connection and AI features being available to the account, and a trial account limits how long the application stays available. At the time of the screenshots the trial showed 13 days remaining.

1.6 Organisation of the Report

Chapter 2 studies the problem area and the related sources and states the proposed system. Chapter 3 lists the requirements and the main use cases. Chapter 4 presents the architecture, data model and workflow design. Chapter 5 describes the implementation, Chapter 6 testing, Chapter 7 deployment, security and governance, and Chapter 8 the results and a discussion of the low-code approach. Chapter 9 concludes and lists future work. The appendices contain the feature checklist, course-outcome mapping, code listings, user manual, evaluation rubric and viva-voce record sheet.

1.7 E-Waste Management Concepts

Three ideas from e-waste management shaped the application. The first is segregation by risk: an item that contains hazardous material, for example the printer and monitor entries in the application, needs more careful handling than an item that does not. The second is age: an item that has been stored for several years is more likely to be obsolete and is less likely to be reused. The third is condition: an item recorded as Working may be redeployed, while one recorded as Not Working or Damaged can only be recycled.

These ideas are the four factors of the priority calculation: quantity, age, hazardous material and condition. Quantity is added because a larger consignment takes more effort to collect. The report does not claim that these four factors are the only valid ones; they are the factors chosen for this application.

1.8 From Collection to Recycling: The Life of an E-Waste Item

Table 1.1 follows one item through the application. The same sequence is used in the walkthrough in Section 5.12.

Table 1.1: Life of an E-Waste Item in EcoCycle

Step	Stage	What Happens in EcoCycle	Where It Is Recorded
1	Identification	A user enters the item with its category, quantity, age, hazardous flag, condition and location.	E-Waste Entry form
2	Prioritisation	A priority score and level are calculated; collection status is set to Pending and recycling status to Not Processed.	Priority Score, Priority Level, Collection Status, Recycling Status
3	Alert	If the level is High, an e-mail notification is sent.	Workflow action
4	Collection request	A requester asks for collection, with a preferred date and time; the details are sent to Zoho CRM as a lead.	Collection Request form; Zoho CRM Leads
5	Collection	The request status moves through Pending, Scheduled and Collected.	Request Status, Collection Completed Date
6	Recycling	The item is recorded with its recycling method and recycler and its status is followed.	Recycling Tracking form
7	Monitoring	Reports, the dashboard and the chat agent show the position.	Reports, EcoCycle Dashboard, AI Agents

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The project description does not describe an existing way of recording e-waste, so the report does not assume one. What can be stated from the project itself is that the application replaces manual judgement of urgency with a calculated score, and replaces separate lists of items, requests and recycling jobs with three linked records and a dashboard.

Figure 2.1 shows the proposed flow of one e-waste entry through the application, from submission to the priority level and the alert decision.

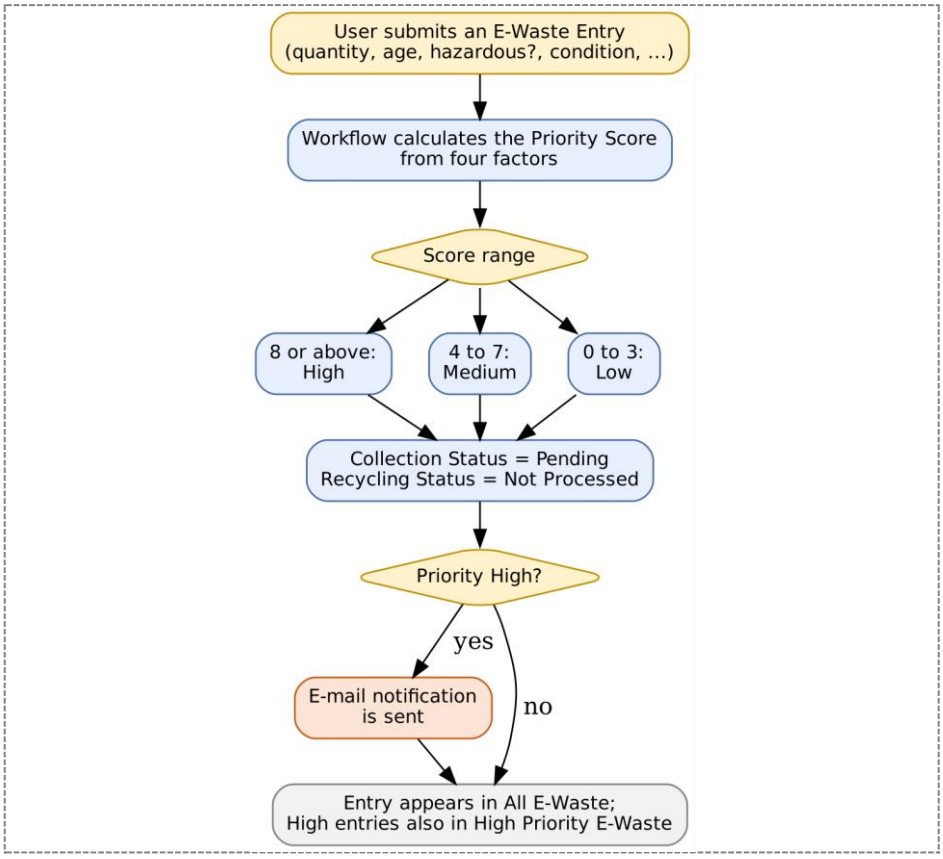


Figure 2.1: Proposed Flow for a Submitted E-Waste Entry

2.2 Limitations of the Evidence Available

- The weight given to each of the four priority factors was not supplied; only the score ranges for High, Medium and Low were given.
- The Deluge source of the workflows was not supplied, and no workflow screen was captured.

- The recipient and text of the high-priority e-mail were not recorded.
- The built-in Zoho Creator REST API integration is a planned enhancement and was not implemented; the Custom API EcoCycle Summary API exists but its response and purpose were not captured.
- How the Sentiment field is filled was not recorded.
- Only one chat-agent exchange was captured.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the traditional route with the route used for this application. The comparison is qualitative; no effort or performance measurements were taken.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed System

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Data model and screens	Tables, migrations and form pages are coded	Each form creates its table and data-entry page
Business rules	Written in a back-end language and deployed	Written as form workflows in Deluge and run on form submission
Reports and dashboard	Query code, a charting library and endpoints	Reports and a page with chart components
CRM integration	API client code, token handling and a server	A connection to Zoho CRM configured under Microservices
AI features	Model hosting and integration code	Chat agent configured on the platform [4], [5]
Control	Full control of every behaviour	Limited to what the platform offers

2.4 Literature Survey

Table 2.2 lists the sources consulted. Only sources that could be verified are included, and the summaries are limited to what the sources themselves state.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. Bock and U. Frank [1], 2021	Low-Code Platform	Discussion paper on the low-code platform concept; its keywords include citizen developer and visual development.	Frames the application as a build on a low-code platform.
2	A. Sahay et al. [2], 2020	Supporting the understanding and comparison of low-code development platforms	Technical survey of eight low-code platforms using a conceptual comparison framework; platforms are heterogeneous.	Supports checking a platform's features against requirements before choosing it.
3	K. Baldé et al. (UNITAR and ITU) [3], 2024	The Global E-waste Monitor 2024	Reports 62 million tonnes of e-waste generated in 2022, of which 22.3% was documented as formally collected and recycled.	Motivates keeping a record of e-waste from entry to recycling.
4	Zoho Corporation [4]	Understanding AI Agent (Zoho Creator Help)	Describes AI agents in Zoho Creator.	Background to the chat agent used in the application.
5	Zoho Corporation [5]	Understanding and configuring chat agent (Zoho Creator Help)	Describes how the chat agent is configured.	Basis of the EcoCycle Data Assistant.

2.5 Emerging Trends in Low-Code Development

Two themes in the sources shaped the design. First, low-code platforms are presented as a way to build business applications visually and quickly [1], while differing in the features they offer [2], so each requirement was checked against what Zoho Creator provides. Second, the platform now includes AI agents and chat agents [4], [5], which allowed a data assistant to be added without leaving the platform. The report does not claim any benefit beyond what the application itself shows.

2.6 Proposed System

The proposed system is EcoCycle. Users register e-waste, request collection and track recycling; the application calculates priority, initialises statuses, rejects past collection

dates, e-mails an alert for high-priority items, creates a CRM lead for each request, displays the position on a dashboard and lets users query the data through a read-only chat agent.

2.7 Technology Study

Table 2.3 lists the technologies used and the role of each. All of them are components of Zoho Creator or Zoho services; no outside library was used.

Table 2.3: Technologies Used and Their Role

Technology	Category	Role in EcoCycle
Zoho Creator	Low-code platform	Hosts the forms, reports, dashboard, workflows and menu
Deluge	Scripting language	Priority calculation, status defaults, date check and e-mail alert
Zoho CRM	Customer-relationship service	Receives a lead for each collection request through a connection
Zia AI chat agent	AI	EcoCycle Data Assistant; answers questions with read-only access
Custom API	Integration	EcoCycle Summary API: a GET endpoint secured with OAuth2
Creator reports and pages	Presentation	Four reports and the EcoCycle Dashboard
E-mail notification	Messaging	Alert for high-priority e-waste

2.8 Feasibility Study

Technical feasibility: every feature maps to a component that Zoho Creator provides, as Table 2.3 shows. Economic feasibility: the application was built on a trial account, so no licence cost was incurred; continued use after the trial would need a paid plan. Operational feasibility: the forms use dropdowns, a date picker and a phone field, so a user can record an item without training. Schedule feasibility: no schedule data was recorded and none is claimed.

2.9 Risk Analysis

Table 2.4: Project Risks and Mitigation

Risk	Effect	Mitigation
Trial expires	Application becomes unavailable	Upgrade the plan or move the application to a new account before the trial ends
Priority weights not documented	Scores cannot be reproduced or audited	Document the weights in Appendix C and in the workflow description
CRM connection fails or is revoked	Leads are not created	Check the connection status under Microservices; re-authorise when needed
Wrong or duplicate data entered	Reports and charts mislead	Add validation and duplicate checks (Section 4.11)
AI answers are wrong	Users act on incorrect information	The chat page advises users to review and verify AI-generated responses

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

The project description names no user roles. Table 3.1 therefore lists the parties that appear in the application itself, and the roles the application should define are left for future work.

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Application user	The person who enters e-waste, requests collection and reads the reports; no named roles were defined	Enter e-waste, submit collection requests, update recycling, read the dashboard and use the chat agent
Requester	Appears in the Requester Name field of a collection request	Asks for e-waste to be collected from a location on a preferred date and time
Recycler	Appears in the Recycler Name field of a recycling record	Recycles the collected e-waste
Zia chat agent	EcoCycle Data Assistant with read-only access	Answers questions about the stored data
External client	Any system that calls the EcoCycle Summary API with OAuth2	Reads a summary of the application data

3.2 Functional Requirements

Table 3.2 lists the functional requirements derived from the project objectives and modules. In the Status column, Built means the item is visible in a screenshot, Reported means it is stated in the project description but not shown, and Planned means it is a future enhancement that was not implemented.

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Status
FR-01	Store e-waste details: name, category, quantity, age, hazardous flag, condition, location and contact details.	User	Built
FR-02	Calculate a priority score from quantity, age, hazardous material and condition.	System	Reported

FR ID	Requirement	Role(s)	Status
FR-03	Classify priority as High (8 or above), Medium (4 to 7) or Low (0 to 3).	System	Built
FR-04	Set collection status to Pending and recycling status to Not Processed on a new entry.	System	Reported
FR-05	Reject a collection request with a past preferred date.	System	Reported
FR-06	Send an e-mail notification for high-priority e-waste.	System	Reported
FR-07	Capture collection requests with requester, location, category, quantity, preferred date and time.	User	Built
FR-08	Track recycling with status, method, recycler name and remarks.	User	Built
FR-09	Provide the All E-Waste, High Priority E-Waste, Collection Requests and Recycling Tracking reports.	User	Built
FR-10	Show a priority pie chart, a collection-status bar chart and a recycling-status bar chart on a dashboard.	User	Built
FR-11	Create a Zoho CRM lead when a collection request is submitted.	System	Built
FR-12	Provide a chat agent that answers questions on the data with read-only access.	User	Built
FR-13	Integrate with the Zoho Creator REST API.	Developer	Planned
FR-14	Limit access by role and profile.	Admin	Not shown
FR-15	Show a read-only Sentiment value for the remarks of a collection request.	System	Built
FR-16	Expose a summary of application data through a Custom API protected by OAuth2.	External client	Built

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement and Evidence
NFR-01	Usability	Forms use dropdowns, radio buttons, date pickers and a phone field with country code, as seen in the form screenshots.
NFR-02	Responsiveness	The application preview offers desktop, tablet and mobile views; behaviour on those views was not checked.
NFR-03	Integration security	The CRM connection is held under Microservices and shows the Connected status.
NFR-04	Safety of AI use	The chat agent has read-only access and the page asks users to review and verify AI-generated responses.
NFR-05	Availability	The application is available while the trial lasts; 13 days remained when the screenshots were taken.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (India data centre)
Scripting	Deluge
CRM	Zoho CRM (Leads module)
AI	Zia chat agent: EcoCycle Data Assistant
Client device	Any computer with a web browser and Internet access
Test data	Four e-waste entries, four collection requests and three recycling records

3.5 Use Case Model

Figure 3.1 shows the actors and the main use cases. Table 3.5 describes the central use case.

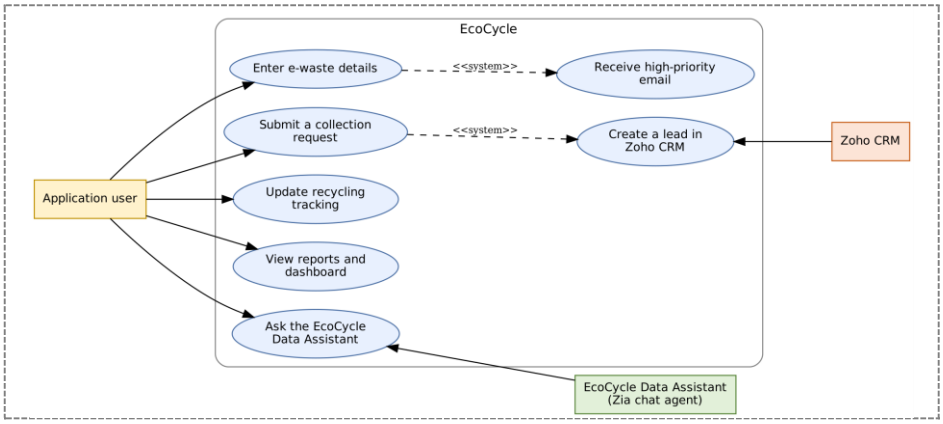


Figure 3.1: Use Case Diagram of EcoCycle

Table 3.5: Use Case Description – Record an E-Waste Entry

Use Case ID	UC-01
Use Case Name	Record an e-waste entry and obtain its priority
Actors	Application user (primary)
Pre-conditions	The user can open the EcoCycle application.
Main Flow	1. The user opens E-Waste Entry. 2. The user enters name, category, quantity, age, hazardous flag, condition, location and contact details. 3. The user submits the form. 4. The workflow calculates the priority score and level. 5. Collection status is set to Pending and recycling status to Not Processed. 6. If the level is High, an e-mail notification is sent.
Post-conditions	The entry appears in All E-Waste, and in High Priority E-Waste when its level is High.
Exceptions	A mandatory field left blank prevents submission.

3.6 Module-wise Functional Specification

Table 3.6: Module-wise Functional Specification

Module	Function	Inputs / Outputs
E-Waste Entry	Capture and prioritise e-waste	In: item details. Out: score, level, statuses
Collection Request	Capture requests for collection and send the requester to Zoho CRM	In: requester, location, category, quantity, date, time. Out: request record and CRM lead
Recycling Tracking	Track recycling activity	In: item, collection date, status, method, recycler. Out: tracking record
Reports	List records with search and sort	Four reports
EcoCycle Dashboard	Summarise priority, request status and recycling status	Three charts and three embedded reports
AI Agents	Answer questions in plain language	In: a question. Out: a table or text answer

3.7 Data Dictionary

Table 3.7 lists the fields seen in the forms. The Mandatory column reflects the red asterisk shown next to a field label. Field link names and the options of dropdown fields (other than those seen in data) are not visible in the screenshots and are not listed.

Table 3.7: Data Dictionary

Form	Field Label	Type	Mandatory	Remarks
E-Waste Entry	E-Waste Name	Single line	Yes	Free text
	Category	Dropdown	Yes	Values seen: Laptop, Computer, Printer, Monitor
	Quantity	Number	Yes	Placeholder #######
	Age(years)	Number	Yes	Placeholder #######
	Contains Hazardous Material?	Radio	Yes	Yes / No
	Condition	Dropdown	No	Values seen: Working, Not Working, Damaged
	Collection Location	Single line	Yes	Block names seen
	User Name, Email	Single line, Email	Yes	—

Form	Field Label	Type	Mandatory	Remarks
	Phone Number	Phone	No	Country code +91 by default
	Priority Score	Number	No	Calculated by workflow
	Priority Level	Dropdown	No	High, Medium, Low
	Collection Status	Dropdown	No	Pending at creation
	Recycling Status	Dropdown	No	Not Processed at creation
Collection Request	Requester Name, Email	Single line, Email	Yes	—
	Phone Number	Phone	Yes	Country code +91 by default
	Collection Location	Single line	Yes	—
	E-Waste Category	Dropdown	Yes	Computer, Printer, Monitor, Laptop seen
	Quantity	Number	Yes	—
	Preferred Collection Date	Date	Yes	dd-MMM-yyyy
	Preferred Time	Dropdown	Yes	Value seen: Morning
	Request Status	Dropdown	No	Pending, Scheduled, Collected seen on chart
	Collection Completed Date	Date	No	dd-MMM-yyyy
	Remarks	Multi line	No	—
	Sentiment	Read-only (greyed)	No	Shows Positive for a praising remark; how it is set was not recorded
Recycling Tracking	E-Waste Name	Single line	Yes	—
	Category	Dropdown	Yes	—
	Quantity	Number	Yes	—
	Collection Date	Date	Yes	dd-MMM-yyyy
	Recycling Status	Dropdown	Yes	Default Not Processed
	Recycling Method	Dropdown	No	Options not shown

Form	Field Label	Type	Mandatory	Remarks
	Recycler Name	Single line	No	—
	Remarks	Multi line	No	—

3.8 Additional Use Cases

Table 3.8: Use Case Description – Submit a Collection Request

Use Case ID	UC-02
Use Case Name	Request collection of e-waste
Actors	Application user; Zoho CRM (secondary)
Pre-conditions	The Zoho CRM connection is in the Connected state.
Main Flow	1. The user opens Collection Request. 2. The user enters requester details, location, category, quantity, preferred date and time. 3. The user submits the form. 4. The date check accepts a date that is not in the past. 5. The request status is set and the requester is sent to Zoho CRM, where a lead is created.
Post-conditions	The request appears in the Collection Requests report and a lead appears in Zoho CRM.
Exceptions	A past preferred date is rejected.

Table 3.9: Use Case Description – Track Recycling

Use Case ID	UC-03
Use Case Name	Record and follow recycling
Actors	Application user
Main Flow	1. The user opens Recycling Tracking. 2. The user enters the item name, category, quantity and collection date. 3. The recycling status defaults to Not Processed; the user may enter a method, recycler name and remarks. 4. The user submits the form.
Post-conditions	The record appears in the Recycling Tracking Report and in the Recycling Status chart.

Table 3.10: Use Case Description – Ask the EcoCycle Data Assistant

Use Case ID	UC-04
Use Case Name	Ask a question about the data
Actors	Application user; Zia chat agent (secondary)
Main Flow	1. The user opens AI Agents. 2. The user types a question, for example "give the e-waste entries with high priority". 3. The agent reads the data and returns a table with a link to the source report.
Post-conditions	No record is changed, because the agent has read-only access.

3.9 Assumptions and Constraints

- The weights of the four priority factors are not part of the evidence; Appendix C gives a reference scheme that is consistent with the stated thresholds and with the levels observed in the test data.
- Details such as the e-mail recipient and the CRM field mapping beyond those visible in the Leads screenshot are not recorded.
- The application ran in one Zoho Creator instance on a trial account.
- The REST API integration is planned and is not part of the delivered application.

3.10 Requirements Traceability

Table 3.11 traces each functional requirement to the module that supplies it and to the place where it is evidenced or tested.

Table 3.11: Requirements Traceability Matrix

FR ID	Module	Evidence / Test	Report Section	Test Case
FR-01	E-Waste Entry	Figures 5.1, 5.2, 5.6	5.2	TC-14
FR-02	Priority workflow	Reference code C.1	5.5	TC-09
FR-03	Priority workflow	Pie chart, High Priority report	5.4	TC-01, TC-11 to TC-13
FR-04	Priority workflow	Recycling default, Pending chart bar	5.6	TC-06, TC-09
FR-05	Past-date check	Reference code C.2	5.6	TC-07, TC-15, TC-16
FR-06	Alert workflow	Reference code C.3	5.6	TC-08
FR-07	Collection Request	Figures 5.3, 5.4, 5.7	5.2	TC-17

FR ID	Module	Evidence / Test	Report Section	Test Case
FR-08	Recycling Tracking	Figures 5.5, 5.8	5.2	TC-06
FR-09	Reports	Figures 5.6 to 5.8, 5.12	5.3	TC-02
FR-10	Dashboard	Figures 5.9 to 5.11	5.4	TC-03, TC-19
FR-11	CRM connection	Figures 5.13 to 5.15	5.7	TC-04, TC-17, TC-18
FR-12	Chat agent	Figure 5.16	5.9	TC-05, TC-20, TC-21
FR-13	REST API	Not implemented	9.2	–
FR-14	Roles	Not shown	4.6	–
FR-15	Collection Request	Figure 5.17	5.16	TC-23
FR-16	Custom API	Figure 5.18	5.17	TC-24

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The application runs entirely on Zoho Creator (Figure 4.1). Users reach it through a browser. Forms, reports and the dashboard form the presentation layer; form workflows written in Deluge form the logic layer; Creator's cloud database holds the data. A connection to Zoho CRM sends each collection request's requester details to the Leads module, and the Zia chat agent reads the application data.

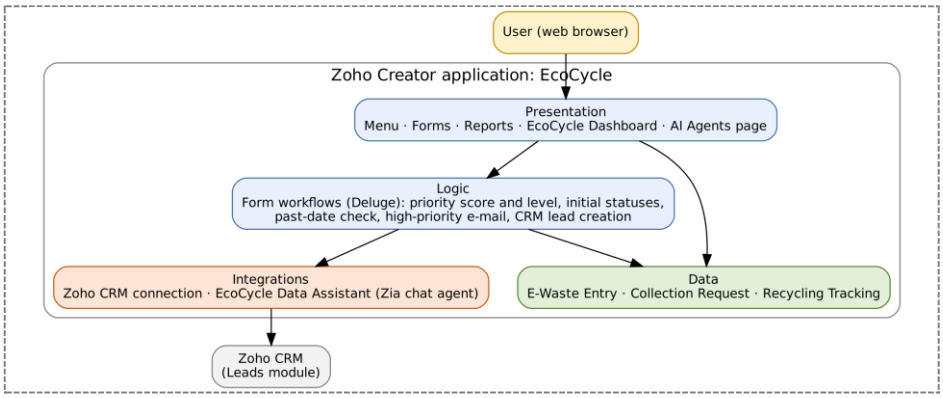


Figure 4.1: System Architecture of EcoCycle

4.2 Data Model

The data model has three data forms (Figure 4.2). No lookup field is visible in any of them: the Recycling Tracking form repeats the e-waste name and category as plain fields, and the Collection Request form repeats the category. The forms are therefore related by the values they hold, not by lookups. Table 4.1 lists the forms and the CRM relationship.

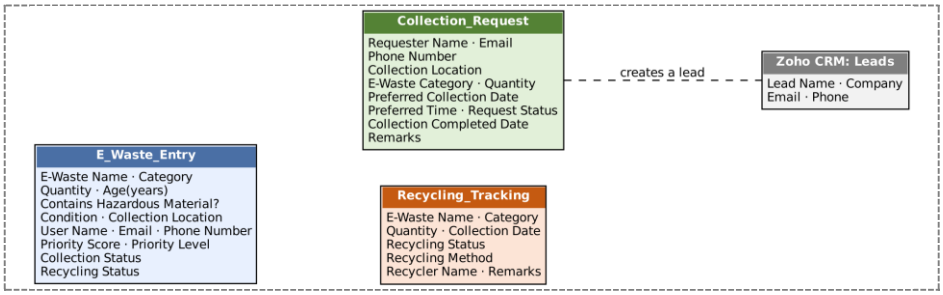


Figure 4.2: Data Model of EcoCycle

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
E-Waste Entry	E-Waste Name, Category, Quantity, Age, Hazardous, Condition, Priority Score and Level	Recycling Tracking; Collection Request	By value (no lookup seen)	Repeated name and category fields
Collection Request	Requester Name, Location, Category, Quantity, Preferred Date and Time, Request Status	Zoho CRM Leads	One request, one lead	Connection to Zoho CRM
Recycling Tracking	E-Waste Name, Category, Quantity, Collection Date, Recycling Status, Method, Recycler	E-Waste Entry	By value (no lookup seen)	Repeated name and category fields

4.3 Form Design

The three forms are single-page forms with the labels on the left and the inputs on the right (Figures 5.1 to 5.5). Mandatory fields carry a red asterisk. Dropdown fields show -Select- until a value is chosen; the Recycling Status field of the Recycling Tracking form opens with Not Processed already selected. The phone fields offer a country-code selector with +91 as the default. Table 3.7 gives the field specification.

4.4 Process and Workflow Design

Figure 2.1 showed the flow of an entry. Table 4.2 states the rules. The score ranges are those given in the project description; the weights of the four factors are not part of the evidence (see Appendix C for a reference scheme).

Table 4.2: Priority and Status Rules

Rule	Condition	Result
Priority score	Calculated from quantity, age, hazardous material and condition	Stored in Priority Score
High	Score 8 or above	Priority Level = High
Medium	Score 4 to 7	Priority Level = Medium
Low	Score 0 to 3	Priority Level = Low
Collection status	New entry	Pending
Recycling status	New entry	Not Processed
Past-date check	Preferred collection date earlier than today	Request rejected
High-priority alert	Priority Level = High	E-mail notification sent

4.5 User Interface and Navigation Design

The application menu (Figure 4.3) has E-waste Entries, App Preferences, Collection Requests, Recycling Tracking, EcoCycle Dashboard and AI Agents. E-waste Entries, Collection Requests and Recycling Tracking each expand to show their form and their reports. App Preferences opens the Manage Integrations panel, which lists the Zoho CRM connection.

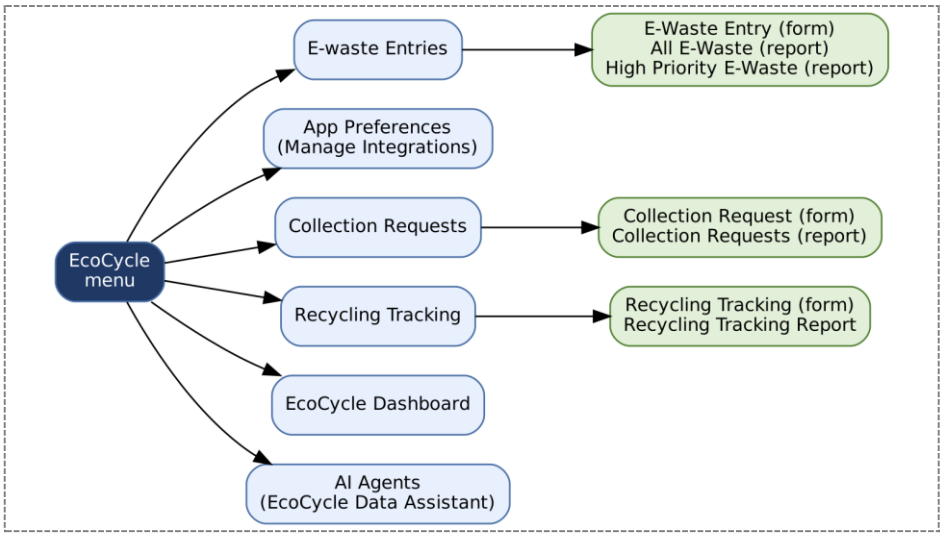


Figure 4.3: Navigation Structure of the Application

4.6 Access Control Design

No roles, permission profiles or record-level access rules were described, and none is visible in the screenshots. The application therefore relies on the default access of the

Zoho Creator account that owns it. Defining roles for staff, collectors and viewers is listed as future work in Chapter 9.

4.7 Detailed Design of the Workflows

The project description lists five automations. Tables 4.3 to 4.5 give the design of the three that carry the main rules. They describe what each workflow does, not its source.

Table 4.3: Design of the Priority Calculation Workflow

Item	Design
Form	E-Waste Entry
Trigger	Successful form submission (create)
Inputs	Quantity, Age(years), Contains Hazardous Material?, Condition
Processing	Add a number of points for each of the four factors to obtain the score; compare the score with 8 and 4
Outputs	Priority Score, Priority Level; Collection Status = Pending; Recycling Status = Not Processed
Reference	Appendix C, Listing C.1

Table 4.4: Design of the Past-Date Validation

Item	Design
Form	Collection Request
Trigger	Form validation, before the record is saved
Condition	Preferred Collection Date is earlier than the current date
Action	Show a message and cancel the submission
Reference	Appendix C, Listing C.2

Table 4.5: Design of the High-Priority E-mail and CRM Lead Workflows

Item	Design
E-mail alert	E-Waste Entry, successful submission; when Priority Level is High, send an e-mail notification (Listing C.3)
CRM lead	Collection Request, successful submission; create a Lead in Zoho CRM using the requester name, e-mail, phone and collection location (Listing C.4)
Evidence	The Leads list in Zoho CRM shows one lead for each of the four collection requests

4.8 Dashboard Design

Table 4.6: Dashboard Components

Component	Type	Source	Shows
E-Waste Priority Distribution	Pie chart	E-Waste Entry	Share of High, Medium and Low priority entries
Collection Request Status	Bar chart	Collection Request	Number of requests that are Collected, Scheduled and Pending
Recycling Status	Bar chart	Recycling Tracking	Number of records by recycling status
High Priority E-Waste	Embedded report	High Priority E-Waste report	High-priority entries
Collection Requests	Embedded report	Collection Requests report	All requests
Recycling Tracking	Embedded report	Recycling Tracking Report	All recycling records

4.9 Integration Design – Zoho CRM

Table 4.7 shows how the requester's details in a collection request correspond to the fields of the lead seen in Zoho CRM. The correspondence is read from the screenshots: the lead name equals the requester name, the lead's Company column holds the collection location, and the e-mail address matches. The full mapping configured in the application was not captured.

Table 4.7: Collection Request Fields and Lead Fields

Collection Request Field	Zoho CRM Lead Column	Evidence
Requester Name	Lead Name	monica, swathi, shreya and jenell appear in both
Collection Location	Company	A block, D block, B block and D block appear in both
Email	Email	Addresses match
Phone Number	Phone	Column cut off in the screenshot

4.10 AI Agent Design

Table 4.8: Design of the EcoCycle Data Assistant

Item	Design
Name	EcoCycle Data Assistant
Type	Zia chat agent in the AI Agents page
Questions supported	High-priority e-waste, pending collection requests, hazardous materials, e-waste quantities and recycling statuses
Access	Read-only; records cannot be modified
Output	A table drawn from a report (for example All E Waste) with a link to open it
Safety note	The page states: Review and verify AI-generated responses before proceeding

4.11 Data Validation and Integrity Considerations

Table 4.9: Recommended Validation and Integrity Checks

Check	Reason	Suggested Implementation
Quantity and age not negative	A negative value would distort the priority score	Field validation on E-Waste Entry
Preferred date not in the past	Stated requirement	Form validation on Collection Request
Duplicate request detection	The same requester and location may be entered twice	Check existing records before saving
Completed date after preferred date	A collection cannot be completed before it was requested	Validation on Collection Request
Link recycling records to entries	Names and categories are repeated by hand and may differ	Replace them with a lookup to E-Waste Entry
Recycler name when status is not Not Processed	A processed record without a recycler is incomplete	Conditional mandatory rule

4.12 Screen Specifications

Tables 4.10 to 4.12 describe each data-entry screen as it appears in the live application.

Table 4.10: Screen Specification – E-Waste Entry

Item	Description
Purpose	Record one e-waste item and obtain its priority
Layout	Single column; labels at the left, inputs at the right; Submit button at the foot
Controls	Single-line boxes, dropdowns, number boxes, Yes/No radio buttons and a phone box with country code
Mandatory fields	E-Waste Name, Category, Quantity, Age(years), Contains Hazardous Material?, Collection Location, User Name, Email
Calculated fields	Priority Score, Priority Level, Collection Status, Recycling Status
Next screens	All E-Waste and High Priority E-Waste reports

Table 4.11: Screen Specification – Collection Request

Item	Description
Purpose	Ask for e-waste to be collected
Layout	Single column with a scrolling form area; Submit button at the foot
Controls	Single-line boxes, phone box, dropdowns, date pickers and a multi-line Remarks box
Mandatory fields	Requester Name, Email, Phone Number, Collection Location, E-Waste Category, Quantity, Preferred Collection Date, Preferred Time
Optional fields	Request Status, Collection Completed Date, Remarks
Read-only field	Sentiment (greyed out; shows Positive for the sample remark)
Side effect	A lead is created in Zoho CRM

Table 4.12: Screen Specification – Recycling Tracking

Item	Description
Purpose	Record and follow recycling of collected e-waste
Layout	Single column; Submit button at the foot
Controls	Single-line boxes, dropdowns, a date picker and a multi-line Remarks box
Mandatory fields	E-Waste Name, Category, Quantity, Collection Date, Recycling Status
Defaults	Recycling Status opens as Not Processed
Optional fields	Recycling Method, Recycler Name, Remarks

4.13 Status Lifecycles

Two status fields follow an item through the application. Table 4.13 lists the values seen in the screenshots and where each is used. Values that were not seen are not listed.

Table 4.13: Status Values Seen

Status Field	Values Seen	Where Used
Collection / Request Status	Pending, Scheduled, Collected	Collection Request Status chart; E-Waste Entry and Collection Request forms
Recycling Status	Not Processed (others not seen)	Recycling Tracking form and report; Recycling Status chart
Priority Level	High, Medium, Low	E-Waste Entry; pie chart; High Priority report
Condition	Working, Not Working, Damaged	E-Waste Entry; All E-Waste report

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application EcoCycle was built in Zoho Creator under the owner account shown in the browser address, with the application link name ecocycle. The header of the live application shows the trial banner (Trial expires in 13 days) together with Share and Edit this application options, and desktop, tablet and mobile view icons. The sidebar lists the six menu items described in Section 4.5.

5.2 Forms and Data Capture

The E-Waste Entry form (Figures 5.1 and 5.2) captures the e-waste name, category, quantity, age in years, whether it contains hazardous material (Yes or No radio buttons), condition, collection location, user name, e-mail, phone number, priority score, priority level, collection status and recycling status. The Collection Request form (Figures 5.3 and 5.4) captures the requester name, e-mail, phone number, collection location, e-waste category, quantity, preferred collection date and time, request status, collection completed date and remarks, and a greyed Sentiment field (Section 5.16). The Recycling Tracking form (Figure 5.5) captures the e-waste name, category, quantity, collection date, recycling status, recycling method, recycler name and remarks.

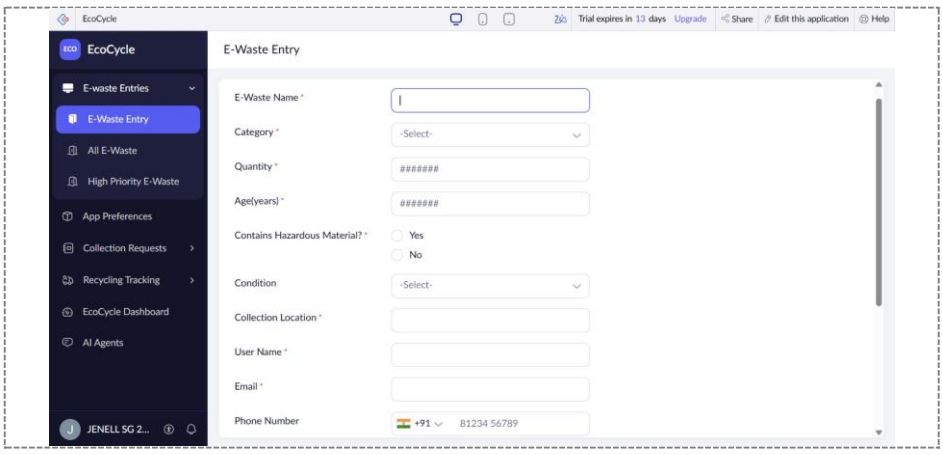


Figure 5.1: E-Waste Entry Form (Upper Part)

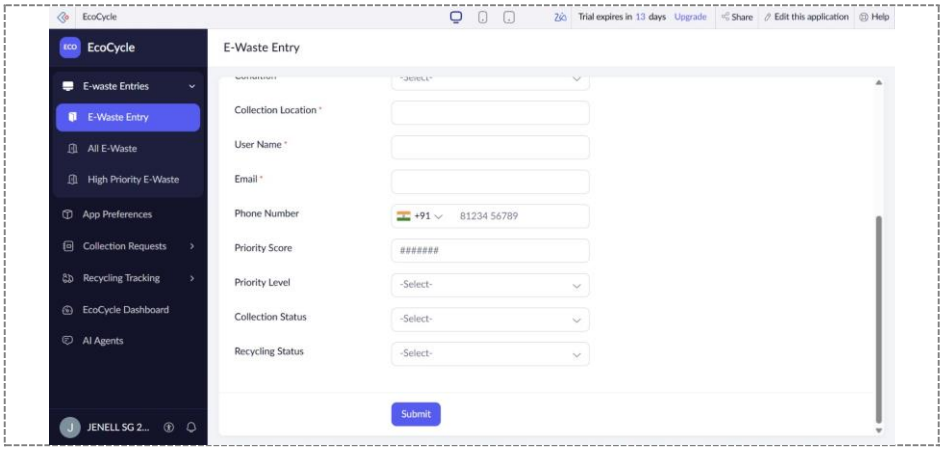


Figure 5.2: E-Waste Entry Form (Lower Part)

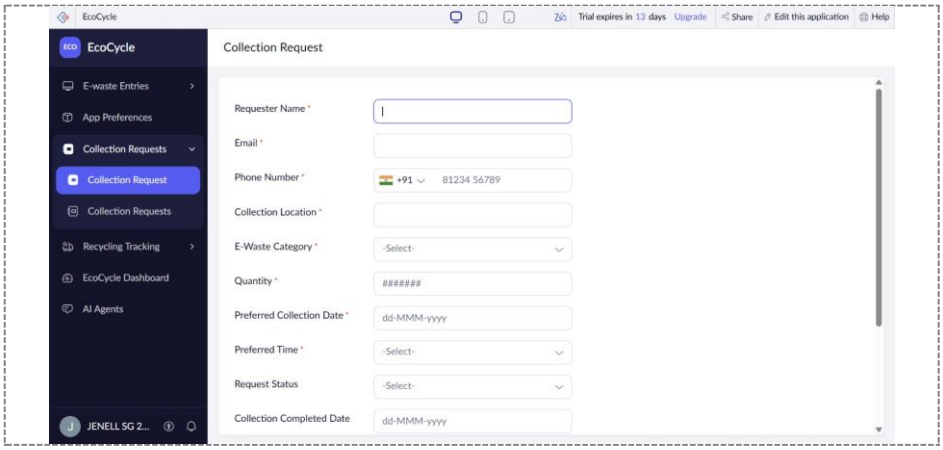


Figure 5.3: Collection Request Form (Upper Part)

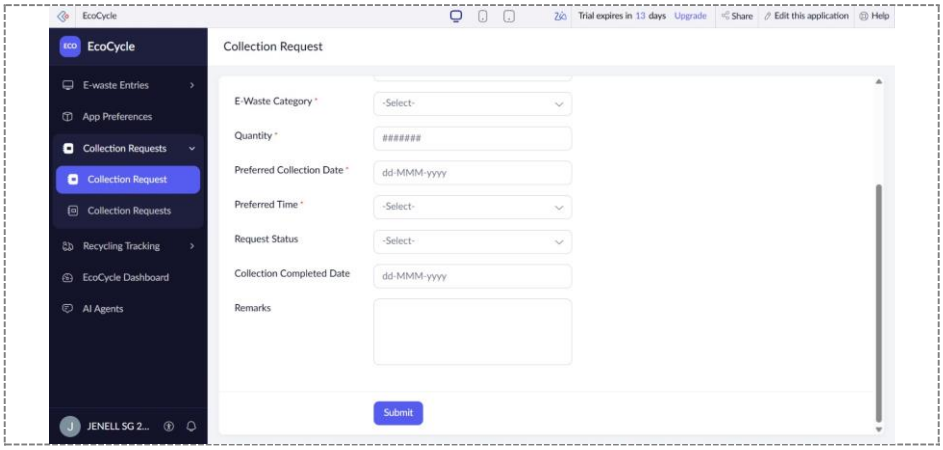


Figure 5.4: Collection Request Form (Lower Part)

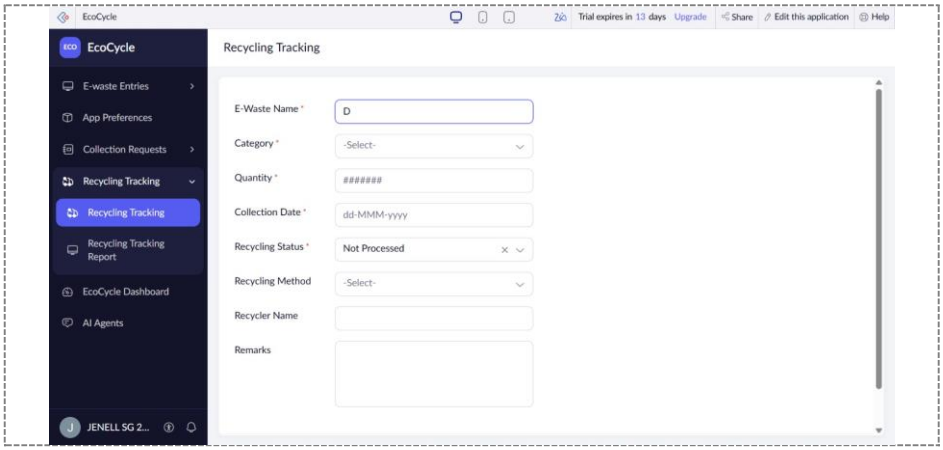


Figure 5.5: Recycling Tracking Form

5.3 Reports

Table 5.1 lists the four reports. Their column sets and the data they hold can be read from the screenshots (Figures 5.6 to 5.8); the filter criteria, grouping and custom actions are not visible, so they are not claimed beyond what the report names and data imply. Each report opens with a record count at the foot, for example Showing 4 of 4.

Table 5.1: Reports Implemented

Report Name	Source Form	Records Shown	Columns Seen	Roles
All E-Waste	E-Waste Entry	4	E-Waste Name, Category, Quantity, Age(years), Contains Hazardous Material?, Condition, Collection Location	Not defined
High Priority E-Waste	E-Waste Entry	2	Same columns; only entries with level High	Not defined
Collection Requests	Collection Request	4	Requester Name, Email, Phone Number, Collection Location, E-Waste Category, Quantity, Preferred date	Not defined
Recycling Tracking Report	Recycling Tracking	3	E-Waste Name, Category, Quantity, Collection Date, Recycling Status, Recycler Name, Remarks	Not defined

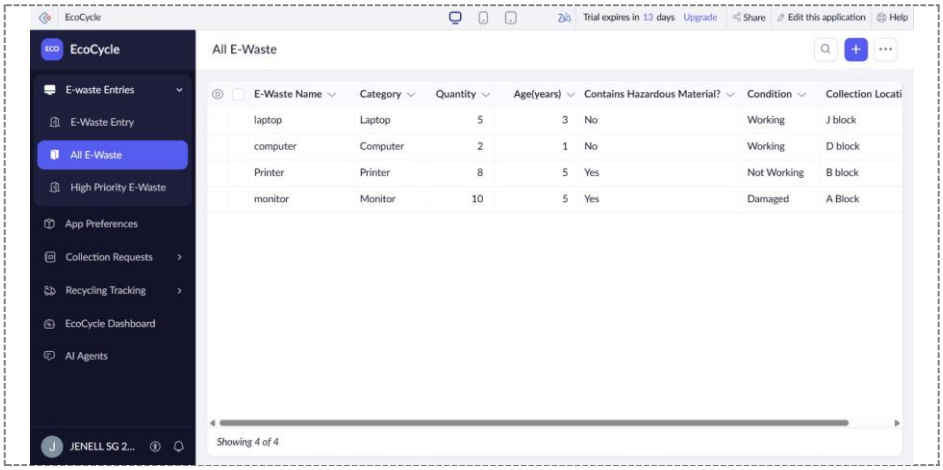


Figure 5.6: All E-Waste Report

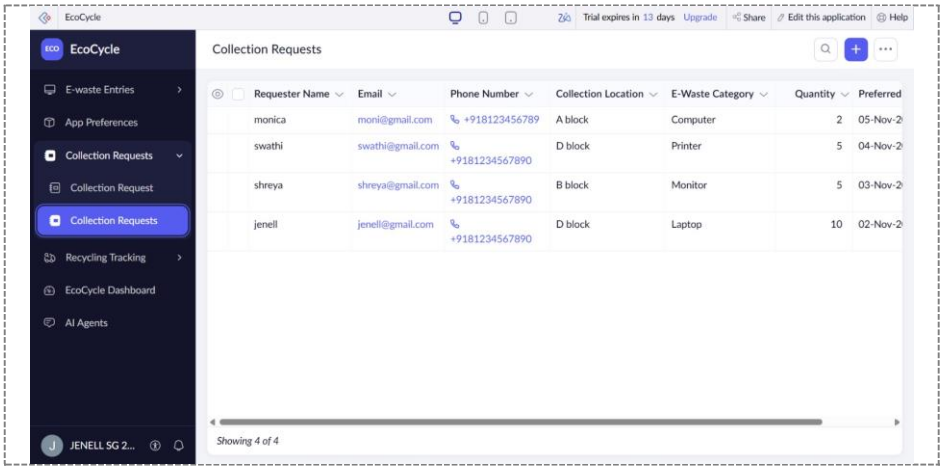


Figure 5.7: Collection Requests Report

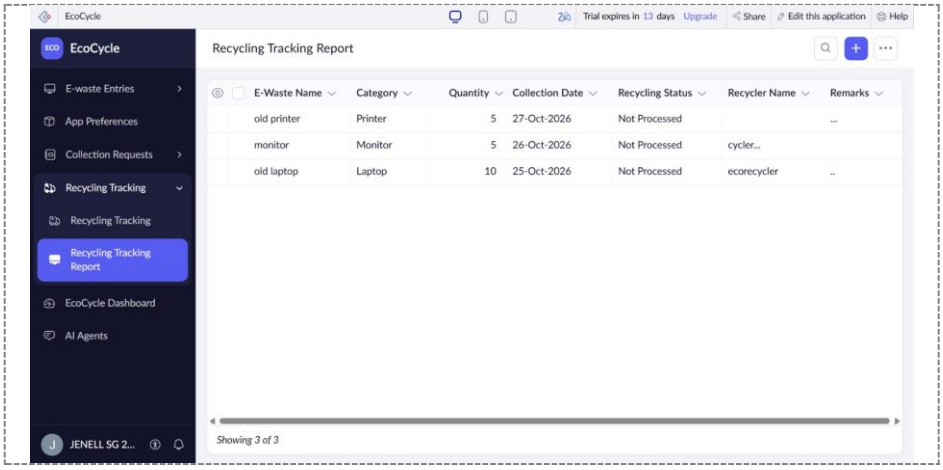


Figure 5.8: Recycling Tracking Report

5.4 Pages and Dashboards

The EcoCycle Dashboard page is headed ECOCYCLE – E-WASTE COLLECTION & RECYCLING DASHBOARD. It stacks three charts and three embedded reports in the order: E-Waste Priority Distribution (a pie chart, Figure 5.9), Collection Request Status (a bar chart, Figure 5.10), Recycling Status (a bar chart, Figure 5.11), High Priority E-Waste (Figure 5.12), Collection Requests and Recycling Tracking.

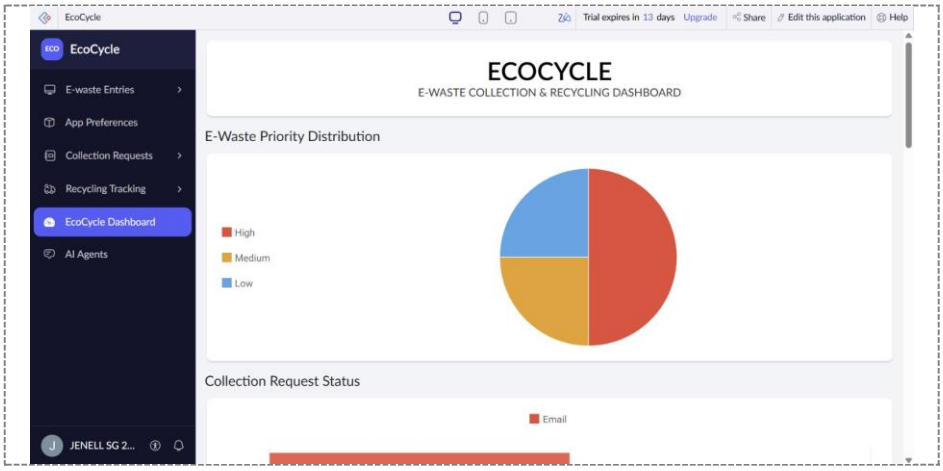


Figure 5.9: EcoCycle Dashboard: Priority Distribution Pie Chart

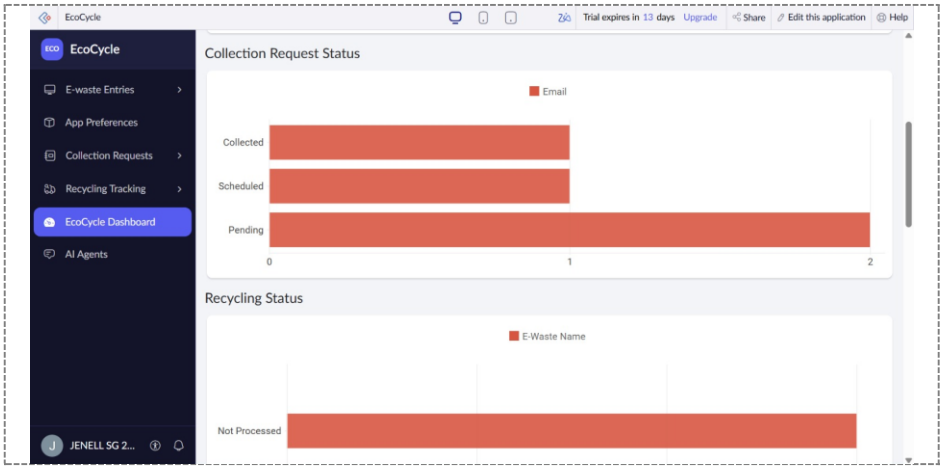


Figure 5.10: EcoCycle Dashboard: Collection Request Status Chart

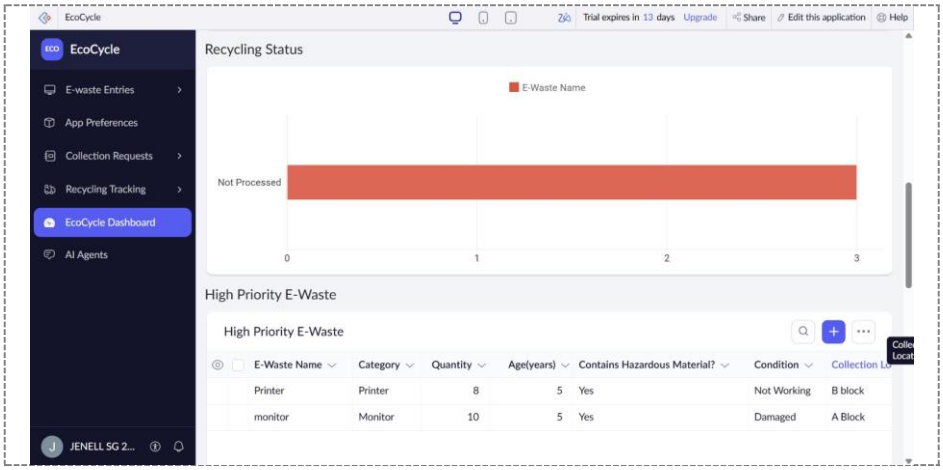


Figure 5.11: EcoCycle Dashboard: Recycling Status Chart and High Priority Report

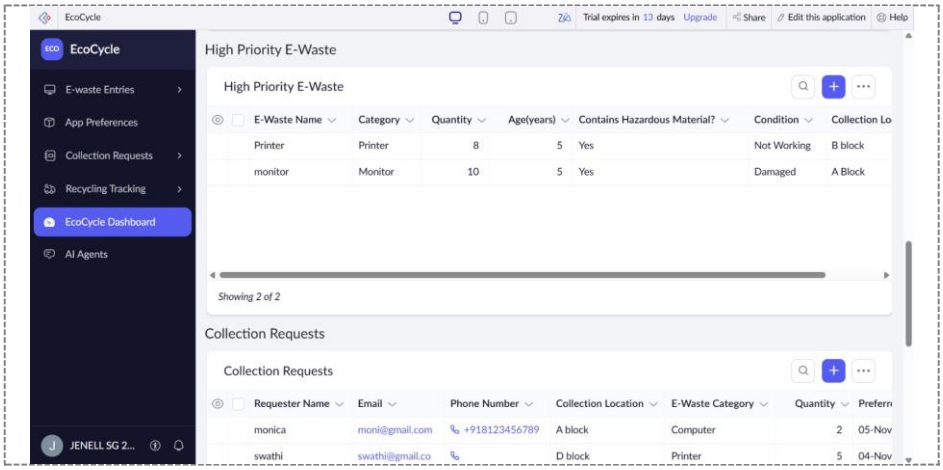


Figure 5.12: EcoCycle Dashboard: High Priority E-Waste and Collection Requests

The pie chart shows half of the entries as High and a quarter each as Medium and Low, which is consistent with two High entries among the four stored entries. The Collection Request Status chart shows one request Collected, one Scheduled and two Pending. The Recycling Status chart shows all three recycling records as Not Processed. The legend of the two bar charts reads Email and E-Waste Name, which are the fields that are counted; a clearer series label is suggested in Table 6.2.

5.5 Deluge Scripting

Table 5.2 lists the automation items stated in the project description. The Deluge source of these items was not supplied and no workflow screen was captured. Appendix C therefore gives reference implementations written from the described design; they are not copies of the application's scripts.

Table 5.2: Deluge Scripts Identified

Script ID	Location / Trigger	Purpose	Source
DS-01	E-Waste Entry, successful submission	Calculate priority score and level; set statuses	Reference implementation, C.1
DS-02	Collection Request, validation	Prevent a past preferred date	Reference implementation, C.2
DS-03	E-Waste Entry, successful submission	E-mail notification for High priority	Reference implementation, C.3
DS-04	Collection Request, successful submission	Create a Zoho CRM lead	Reference implementation, C.4

5.6 Workflows and Automation

Table 5.3 lists the form workflows described for EcoCycle. All are form workflows; no schedule, approval, Blueprint or batch workflow is part of the application. The conditions are those stated in the description.

Table 5.3: Workflows Implemented

Workflow	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Priority calculation	Form	E-Waste Entry – successful submission	Always	Set score, level and statuses	DS-01
Past-date check	Form	Collection Request – validation	Date is in the past	Reject the request	DS-02
High-priority alert	Form	E-Waste Entry – successful submission	Level is High	Send e-mail	DS-03
CRM lead creation	Form	Collection Request – successful submission	Always	Create a lead	DS-04

5.7 Integrations

Table 5.4 lists the integrations. The Connections tab of Microservices holds one connection, Zoho CRM, of type System, in the Connected state (Figure 5.13). The same connection is listed in the Manage Integrations panel opened from App Preferences, as Zoho CRM Account, Connected (Figure 5.14). In Zoho CRM the Leads module lists four leads whose names and e-mail addresses match the four collection requests (Figure 5.15); the Company column holds the collection location.

Table 5.4: Integrations Implemented

Integration	Service	Mechanism	Status Seen	Data Exchanged
Zoho CRM	Leads module	Creator connection (Microservices)	Connected	Requester name, location, e-mail, phone
Chat agent	EcoCycle Data Assistant	Zia chat agent	Working (one exchange)	Question and table of records
Custom API	EcoCycle Summary API	GET endpoint; OAuth2	Ready to invoke	Summary of application data (response not captured)
Creator REST API (built-in)	Zoho Creator	REST	Not implemented	Planned

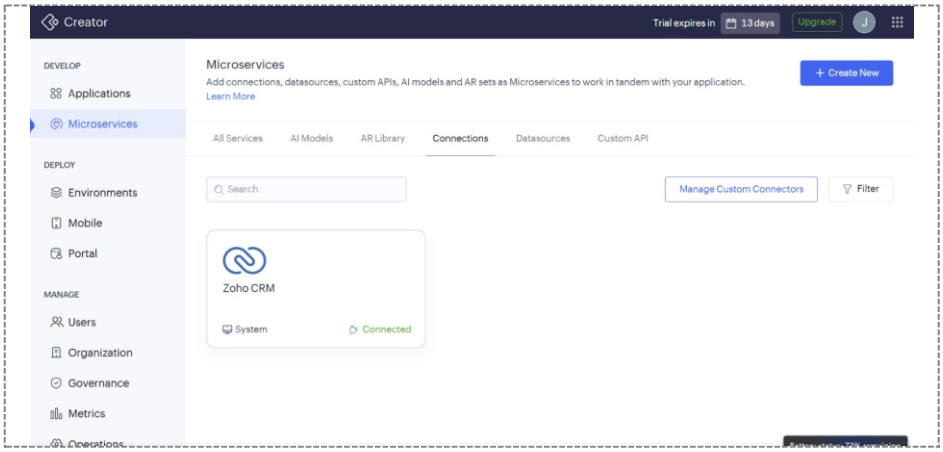


Figure 5.13: Microservices Showing the Zoho CRM Connection

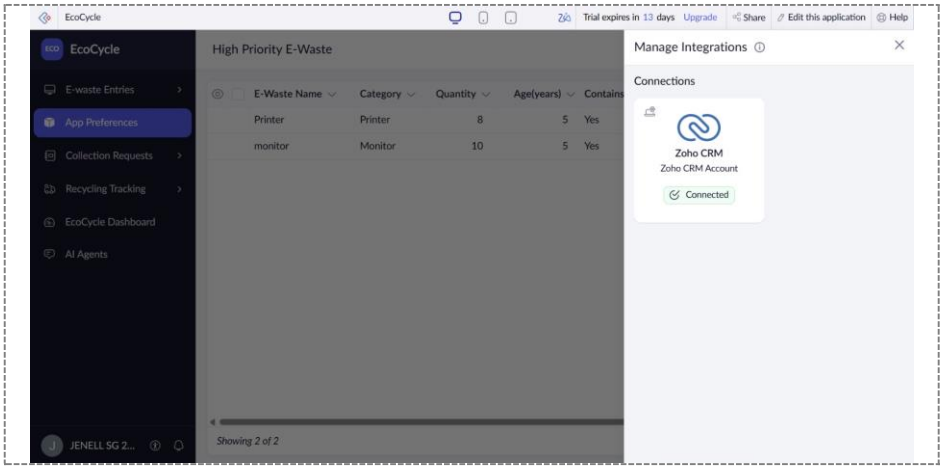


Figure 5.14: Manage Integrations Panel Showing Zoho CRM Connected

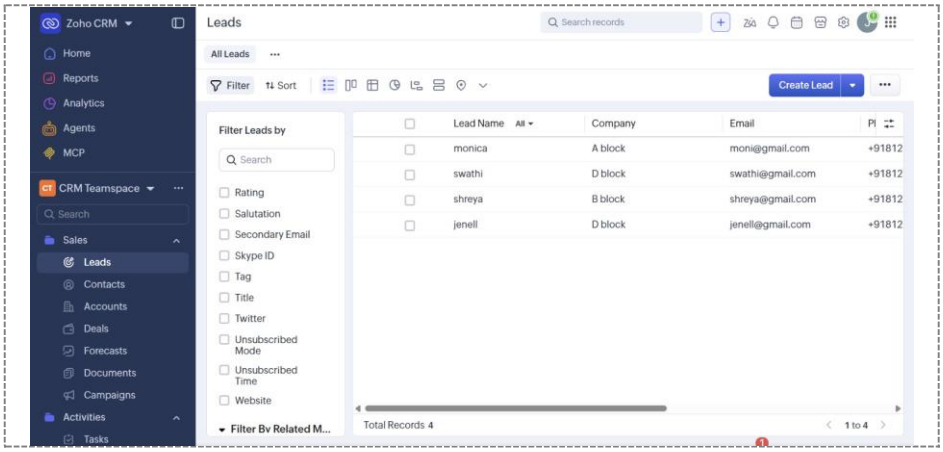


Figure 5.15: Zoho CRM Leads Created from Collection Requests

5.8 User Management and Role-Based Access

No permission profiles, roles or user invitations were described or shown. The header offers a Share option, but its use was not recorded. Roles and permission profiles are listed as future work in Section 9.2.

5.9 AI and Analytics Features

The EcoCycle Data Assistant is reached from the AI Agents menu item. Figure 5.16 shows one exchange: the user asked for the e-waste entries with high priority, and the agent answered with a table taken from the All E Waste report, with the columns Priority Level, E-Waste Name, Category, Quantity and Contains Hazardous Material. The two rows returned (Printer, 8, and monitor, 10, both hazardous) match the High Priority E-Waste report. The page footer reads Review and verify AI-generated responses before proceeding. The agent is described as read-only, so it can retrieve information but not change records.

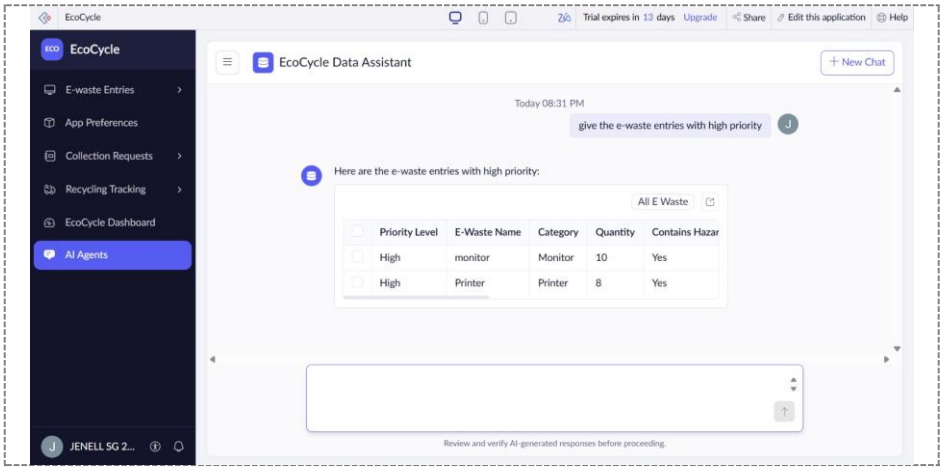


Figure 5.16: EcoCycle Data Assistant Answering a High-Priority Question

5.10 Pro-Code Extension

No pro-code extension is part of EcoCycle. The Custom API of Section 5.17 is a platform feature and is distinct from the built-in REST API. The Zoho Creator REST API integration is planned as a future enhancement to retrieve application data and connect EcoCycle with external applications; it has not been implemented.

5.11 Mobile and Portal Access

The application header shows desktop, tablet and mobile view icons. Use of the Creator mobile app or a customer portal was not described.

5.12 Application Walkthrough

The following sequence follows the screenshots and uses the test data of the application.

- An entry such as the monitor (quantity 10, age 5 years, hazardous, damaged, A block) is saved through E-Waste Entry and appears in All E-Waste (Figure 5.6).
- Because the monitor and the printer are High, both appear in High Priority E-Waste and in the dashboard (Figure 5.12).
- Requests from monica, swathi, shreya and jenell are saved through Collection Request and listed in Collection Requests (Figure 5.7); four leads appear in Zoho CRM (Figure 5.15).
- Recycling records for the old printer, monitor and old laptop are entered through Recycling Tracking and listed with the status Not Processed (Figure 5.8).
- The user opens EcoCycle Dashboard to see the priority split and the two status charts (Figures 5.9 to 5.11), or asks the chat agent a question (Figure 5.16).

5.13 Sample Data and Worked Calculation

Table 5.5 lists the four e-waste entries read from the All E-Waste report. Its last column applies the reference scoring scheme of Appendix C to them. The scheme reproduces the two High levels observed for the printer and the monitor, and it gives one Medium and one Low entry for the laptop and the computer, which matches the pie chart split. The actual stored scores were not visible, so the scores below are those of the reference scheme, not values read from the application.

Table 5.5: Sample E-Waste Entries and Reference Priority

Entry	Qty	Age (yrs)	Haz.	Condition	Location	Ref. Score	Ref. Level
laptop	5	3	No	Working	J block	4	Medium
computer	2	1	No	Working	D block	2	Low
Printer	8	5	Yes	Not Working	B block	10	High
monitor	10	5	Yes	Damaged	A Block	10	High

Table 5.6: Collection Requests and Recycling Records Seen

Record	Category	Quantity	Date	Status
Request: monica	Computer	2	Preferred 05-Nov	Not visible per row
Request: swathi	Printer	5	Preferred 04-Nov	Not visible per row
Request: shreya	Monitor	5	Preferred 03-Nov	Not visible per row
Request: jenell	Laptop	10	Preferred 02-Nov	Not visible per row
Recycling: old printer	Printer	5	Collected 27-Oct-2026	Not Processed
Recycling: monitor	Monitor	5	Collected 26-Oct-2026	Not Processed
Recycling: old laptop	Laptop	10	Collected 25-Oct-2026	Not Processed

5.14 Implementation Status Summary

Table 5.7: Implementation Status Summary

Component	Status	Evidence
Three data forms	Built	Figures 5.1 to 5.5
Four reports	Built	Figures 5.6 to 5.8, 5.12
Dashboard with three charts	Built	Figures 5.9 to 5.12
Priority calculation and levels	Built (levels seen); weights not supplied	Pie chart; High Priority report
Zoho CRM lead creation	Built	Figure 5.15
Chat agent	Built (one exchange captured)	Figure 5.16
Sentiment field	Built (mechanism not recorded)	Figure 5.17
Custom API	Built (response not captured)	Figure 5.18
Workflow source code	Not supplied	Appendix C holds reference code
Creator REST API	Not implemented	Planned

5.15 Report Columns and Their Use

Table 5.8: Use of the Report Columns

Column	Report	Use
Category	All E-Waste; Recycling Tracking	Groups items by type such as Laptop, Printer and Monitor
Quantity	All reports	Number of items; one input to the priority score
Age(years)	All E-Waste	Age of the item; one input to the priority score
Contains Hazardous Material?	All E-Waste	Yes marks the item for careful handling
Condition	All E-Waste	Working, Not Working or Damaged
Collection Location	All E-Waste; Collection Requests	Where the item is to be collected
Preferred Collection Date	Collection Requests	Date requested; must not be in the past
Recycling Status	Recycling Tracking	Progress of recycling
Recycler Name	Recycling Tracking	Who recycles the item

5.16 Sentiment Field in Collection Requests

The lower part of the Collection Request form carries a Remarks box and, below it, a Sentiment field (Figure 5.17). In the screenshot the request is for a quantity of 10 with the preferred date 02-Nov-2026 and the preferred time Morning; the remark says that the collection service was excellent and the team was very helpful, and the Sentiment field shows Positive. The field is greyed out, so it appears to be read-only and filled by the application, not typed by the user. How it is filled (for example by a Zia text-analysis task) was not recorded, so the report states only what the screen shows: a praising remark is labelled Positive.

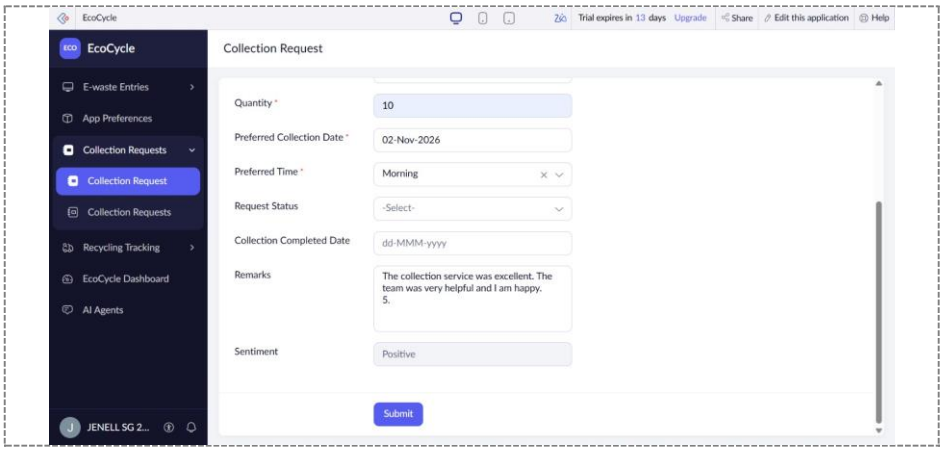


Figure 5.17: Collection Request Form Showing Remarks and the Sentiment Field

5.17 Custom API

The Custom API tab of Microservices lists one API, EcoCycle Summary API (Figure 5.18). It uses the GET method, shows an Endpoint URL that can be copied, is secured with OAuth2 and is in the Ready to invoke state. The name indicates that it returns a summary of the application data, but its function, its parameters and its response were not captured, so no output is described. OAuth2 means that a caller must present a valid access token, which is a stronger control than a public key.

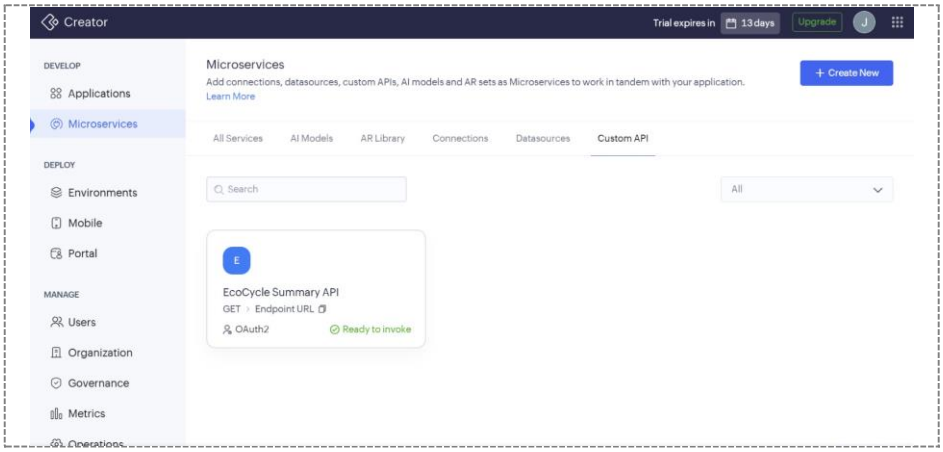


Figure 5.18: Custom API List Showing the EcoCycle Summary API

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

No test plan or UAT record was supplied. Table 6.1 lists the checks that can be confirmed from the screenshots; the remaining checks are planned in Section 6.6. No automated tests were written.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-03	Priority levels shown	Open dashboard pie chart	High, Medium and Low shares shown	High 50%, Medium 25%, Low 25% (Figure 5.9)	Pass (observed)
TC-02	FR-09	High Priority report	Open High Priority E-Waste	Only High entries	Printer and monitor, 2 of 2 (Figure 5.12)	Pass (observed)
TC-03	FR-10	Collection status chart	Open dashboard	Counts by status	Collected 1, Scheduled 1, Pending 2 (Figure 5.10)	Pass (observed)
TC-04	FR-11	CRM lead creation	Open Zoho CRM Leads	One lead per request	4 leads for 4 requests (Figure 5.15)	Pass (observed)
TC-05	FR-12	Chat agent answer	Ask for high-priority entries	Matches the report	2 rows matching the report (Figure 5.16)	Pass (observed)
TC-06	FR-08	Recycling status default	Open Recycling Tracking form	Not Processed preselected	Not Processed shown (Figure 5.5)	Pass (observed)
TC-07	FR-05	Past date rejected	Submit a request dated yesterday	Rejected	Not executed	Not run
TC-08	FR-06	High-priority e-mail	Submit a High entry	E-mail received	Not executed	Not run
TC-23	FR-15	Sentiment shown	Open a request whose remark praises the service	Sentiment shown read-only	Positive, greyed (Figure 5.17)	Pass (observed)

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-24	FR-16	Custom API listed	Open Microservices → Custom API	API ready	EcoCycle Summary API, GET, OAuth2, Ready to invoke (Figure 5.18)	Pass (observed)

6.3 Debugging Deluge Scripts

No Deluge error messages or debugging sessions were supplied, so none is described. The checks in Section 4.11 are intended to prevent the errors most likely in these workflows, such as a blank or negative number reaching the score calculation. Deluge offers the info statement, used in Listing C.4, to write a value to the log while a workflow is tested.

6.4 Defect Log

Table 6.2 lists the issues that are visible in the screenshots. They are observations, not a full defect history.

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
D-01	Collection Location is typed text, and values differ in case (A Block and A block).	Free-text field	Replace with a dropdown or a lookup of locations	Open
D-02	Recycling records have incomplete Recycler Name and Remarks values (for example cycler... and ..).	Test data entered quickly	Replace with meaningful test data	Open
D-03	Chart legends read Email and E-Waste Name.	Legend shows the counted field	Rename the series to Requests and Records	Open
D-04	Recycling records are not linked to entries.	Name and category are retyped	Add a lookup field	Open
D-05	Trial period ends in 13 days.	Trial account	Upgrade or move the application	Open

Bug ID	Description	Root Cause	Fix Applied	Status
D-06	No response from EcoCycle Summary API was captured.	Not tested in the evidence	Call the endpoint with OAuth2 and record the response	Open

6.5 User Acceptance Testing

No user acceptance testing was recorded. The planned tests in Table 6.3 should be run, with at least one other user, before the application is accepted.

6.6 Planned Test Cases

Table 6.3: Planned Test Cases (Not Yet Executed)

TC ID	Scenario	Expected Result	FR Ref.	Status
TC-09	Entry with quantity 10, age 5, hazardous Yes, condition Damaged	Level High; statuses Pending and Not Processed	FR-02, 03, 04	Planned
TC-10	Entry with quantity 2, age 1, hazardous No, condition Working	Level Low	FR-03	Planned
TC-11	Entry with a score of exactly 8	Level High	FR-03	Planned
TC-12	Entry with a score of exactly 4	Level Medium	FR-03	Planned
TC-13	Entry with a score of exactly 3	Level Low	FR-03	Planned
TC-14	Submit an entry without a mandatory field	Submission blocked	FR-01	Planned
TC-15	Request dated today	Accepted	FR-05	Planned
TC-16	Request dated in the past	Rejected with a message	FR-05	Planned
TC-17	Submit a request	Lead appears in Zoho CRM with the same name and e-mail	FR-11	Planned
TC-18	Disconnect the CRM connection and submit a request	Failure is visible and the request is not lost	FR-11	Planned
TC-19	Change a request status to Collected	Dashboard chart updates	FR-10	Planned
TC-20	Ask the agent for pending collection requests	Two requests returned	FR-12	Planned

TC ID	Scenario	Expected Result	FR Ref.	Status
TC-21	Ask the agent to delete a record	Refused; records unchanged	FR-12	Planned
TC-22	Open the application on a mobile view	Forms and reports remain usable	NFR-02	Planned

6.7 Test Data Design

Table 6.4: Test Data Design

Data Set	Purpose	Example
Boundary scores	Check the 3/4 and 7/8 limits	Entries scoring 3, 4, 7 and 8
Hazardous and non-hazardous items	Check the hazardous factor	Printer (Yes) and laptop (No)
Dates	Check the past-date rule	Yesterday, today and tomorrow
Categories	Check every dropdown value	Laptop, Computer, Printer, Monitor
Phone numbers	Check the country-code field	+91 numbers

6.8 Recording Test Results

Each executed test should be recorded with the date, the input used, the result and a screenshot, and any failure should be added to the defect log with its cause and fix. Planned tests keep the status Planned until they are executed.

6.9 Test Coverage Summary

Table 6.5: Test Coverage Summary

Category	Observed	Planned	Not Run
Priority and levels	1	5	0
Reports and dashboard	2	1	0
Validation	0	3	1
Notification	0	0	1
CRM integration	1	2	0
Chat agent	1	2	0
Forms and usability	1	2	0
Sentiment and Custom API	2	0	0

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed and used in a single Zoho Creator instance; development, stage and production environments were not shown. Table 7.1 records the deployment checklist as far as the evidence goes.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	Application created with forms, reports, dashboard and workflows	✓
2	Zoho CRM connection created and Connected	✓
3	Chat agent available in the application	✓
4	Publish through development, stage and production environments	—
5	Roles and permission profiles defined	—
6	Creator REST API integration	—
7	Plan upgraded before the trial ends	—

7.2 Performance Optimisation

No performance measurements were taken, so no figures are reported. Table 7.2 lists the design choices that bear on performance.

Table 7.2: Performance-Related Design Choices

Design Choice	Component	Measured?	Result	Remarks
Dashboard charts read directly from the forms	EcoCycle Dashboard	No	Not available	Load time not measured
Three embedded reports on the dashboard	EcoCycle Dashboard	No	Not available	Each shows only a few records

7.3 Security Features

Table 7.3: Security Controls Identified

Security Control	Where Applied	Threat Mitigated
Read-only access for the chat agent	EcoCycle Data Assistant	Accidental or malicious change of records through chat
OAuth2 on the Custom API	EcoCycle Summary API	Calls without a valid token are refused

Security Control	Where Applied	Threat Mitigated
Zoho CRM connection held in Microservices	CRM integration	Credentials are not stored in scripts
Sign-in to Zoho Creator	Whole application	Anonymous access
Verify-AI-responses notice	AI Agents page	Over-reliance on AI output

Known gaps: no roles or permission profiles were defined, so every user who can open the application sees all records, including requester e-mail addresses and phone numbers; and the application stores personal contact details without a stated retention rule.

7.4 Governance and Administration

The application is owned by one Zoho Creator account. Audit trail, administration settings and data-retention rules were not described. Sections 4.6 and 9.2 list the controls to add.

7.5 Backup and Recovery Procedure

No backup was described. Table 7.4 recommends the components to back up and how to restore them.

Table 7.4: Backup Components and Recovery

Component	Backup	Recovery
Application structure	Export the application	Import into a new account
Records	Export each form's data	Import into the restored forms
Deluge code	Keep copies of the scripts (Appendix C)	Paste into the workflows
CRM connection	Note its name and scope	Create it again and authorise
Chat agent settings	Record the instructions and access level	Configure the agent again

7.6 Risk Register

Table 7.5: Deployment and Operation Risks

Risk	Likelihood	Impact	Response
Trial expires	High	High	Upgrade or move before the date
CRM connection revoked	Medium	Medium	Monitor the connection status
Personal data exposed	Medium	High	Define roles; limit who can see contact details
Priority weights changed without notice	Low	Medium	Keep the weights documented

7.7 Operational Checklist

Table 7.6: Recommended Operational Checklist

S.No.	Check	Frequency
1	Review High Priority E-Waste and pending requests	Daily
2	Confirm the CRM connection shows Connected	Weekly
3	Update recycling statuses	Weekly
4	Check that the dashboard totals match the reports	Monthly
5	Export a backup of the data	Monthly

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The dashboard and the reports are the visible results of the application (Figures 5.6 to 5.12). Table 8.1 lists the values read from them.

Table 8.1: Values Observed in the Application

Item	Value Observed
E-waste entries	4 (laptop, computer, Printer, monitor)
High-priority entries	2 (Printer, monitor)
Priority distribution (pie)	High half, Medium a quarter, Low a quarter
Collection requests	4: Collected 1, Scheduled 1, Pending 2
Recycling records	3, all Not Processed
Zoho CRM leads	4, matching the four collection requests
Chat agent answer	2 high-priority entries, matching the report

8.2 Achievement of Objectives

Table 8.2: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Evidence	Status
Store e-waste entries	E-Waste Entry	Figures 5.1, 5.2, 5.6	Achieved
Priority score and level	Priority workflow	Figures 5.9, 5.12	Levels seen; weights not supplied
Initial statuses	Priority workflow	Figures 5.10, 5.11	Reported
Reject past dates	Past-date check	Not demonstrated	Reported
High-priority e-mail	Alert workflow	Not demonstrated	Reported
Four reports	Reports	Figures 5.6 to 5.8, 5.12	Achieved
Dashboard	EcoCycle Dashboard	Figures 5.9 to 5.12	Achieved
CRM lead creation	CRM connection	Figure 5.15	Achieved
Chat agent	EcoCycle Data Assistant	Figure 5.16	Achieved

8.3 Intended Capability versus Evidence

No description of an earlier system was supplied, so a comparison with one is not made. Table 8.3 compares what the project set out to do with what the evidence shows.

Table 8.3: Intended Capability versus Evidence Available

Capability	Intended	Evidence Available
Priority	Automatic, three levels	Levels visible in the pie chart and report
Statuses	Defaults on a new entry	Not Processed seen on the form and reports
Date check	Past dates rejected	Not demonstrated
E-mail alert	On High priority	Not demonstrated
CRM	Lead per request	Four leads seen
AI	Read-only answers	One answer seen
Custom API	Summary endpoint	Listed, GET, OAuth2, Ready to invoke; response not captured
Built-in REST API	Future	Not implemented

8.4 Limitations of the Low-Code Approach

- A trial account limits how long the application stays available.
- Connections and AI features are configured outside the forms, so they must be recreated if the application is moved.
- Workflow logic is spread over several workflows, which is harder to review without the Deluge source.
- AI output depends on instructions and cannot be guaranteed; the page itself advises users to verify it.
- Free-text fields such as Collection Location produce inconsistent values unless they are replaced by controlled fields.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of EcoCycle is configuration: forms, reports, charts, the menu, the CRM connection and the chat agent. The rules (priority, statuses, date check and alert) sit at the low-code level in short Deluge workflows. No pro-code layer was needed. A pro-code layer, reached through the planned REST API, would become relevant when

EcoCycle must exchange data with external applications, for example a collection-tracking map or an SMS gateway.

8.6 Interpretation of the Dashboard Results

With four entries, half are High, so the high-priority list is as long as half of the stored items and needs action first. Two of four collection requests are Pending, one is Scheduled and one is Collected. All three recycling records are Not Processed, which shows that collection has run ahead of recycling in the test data. The figures are small test data and are not evidence about any real facility.

8.7 Challenges and Lessons

- Choosing the factors of the priority score required deciding which attributes matter most; the weights should be documented.
- Keeping three forms consistent without lookups showed the value of controlled values.
- Connecting to Zoho CRM showed that the connection must be authorised before a workflow can use it.
- The chat agent needed read-only access to be safe to use on live data.

8.8 Recommendations

Table 8.4: Recommendations Arising from the Review

No.	Recommendation	Related Item
1	Document the priority weights in the workflow and in this report	Section 3.9
2	Replace retyped names with lookups to E-Waste Entry	D-04
3	Use a dropdown for Collection Location	D-01
4	Rename the chart legends	D-03
5	Define roles for staff, collectors and viewers	Section 4.6
6	Run the planned tests of Table 6.3	Section 6.6

8.9 Sustainability and Social Relevance

The Global E-waste Monitor 2024 reports that only 22.3 per cent of the 62 million tonnes of e-waste generated in 2022 was documented as formally collected and recycled [3]. A tool such as EcoCycle does not change that figure by itself, but it supports the

step that documentation depends on: recording each item, deciding which items are urgent, and following them to a recycler. The application is small and its data is test data, so no environmental benefit is claimed for it.

8.10 Strengths and Weaknesses

Table 8.5: Strengths and Weaknesses of the Application

Strengths	Weaknesses
One record of waste, requests and recycling	No lookups between the three forms
Automatic priority classification	Weights not documented in the evidence
Reports and dashboard give a quick view	Chart legends are unclear
CRM leads created without retyping	Depends on the connection staying authorised
Read-only chat agent for quick questions	Only one exchange captured; answers must be verified

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

EcoCycle simplifies e-waste collection and recycling management using Zoho Creator. It combines automated priority calculation, workflow automation, reports, a dashboard, CRM integration and AI assistance to make e-waste management more organised and efficient. It is an application that manages e-waste from entry and priority identification to collection tracking and recycling using automation and AI.

The report states plainly what is not shown: the Deluge sources and the priority weights were not supplied (the code given is a reference), the e-mail alert and the date check were not demonstrated, and the Creator REST API is not implemented. Chapter 6 lists these as open items.

9.2 Future Enhancements

- REST API integration, to retrieve application data and connect EcoCycle with external applications.
- Google Maps integration for collection locations.
- Automated collection scheduling.
- SMS notifications.
- QR-code identification of e-waste items.
- Real-time collection tracking.
- Roles and permission profiles, and lookups between the three forms.

9.3 Learning Outcomes and Career Relevance

The project gave practice in modelling forms, expressing a scoring rule and validation as form workflows, building reports and a dashboard, connecting to Zoho CRM and configuring a chat agent. It also showed the value of recording evidence for each feature. These skills apply to roles in low-code development, business-application support and integration.

9.4 Enhancement Roadmap

Table 9.1: Enhancement Roadmap

Phase	Enhancement	Benefit
Short term	Lookups, dropdown locations, roles, planned tests	Cleaner data and access control
Medium term	Collection scheduling, SMS notifications, REST API	Less manual follow-up; external integration
Long term	Google Maps, QR codes, real-time tracking	Route planning and item-level traceability

9.5 Scope of the Claims in this Report

Every statement in this report is tied either to a screenshot of the application or to the project description. Where the evidence is a description only (for example the e-mail alert and the past-date check), the report says Reported. Where it is a reference value (the priority weights and the listings in Appendix C), the report says so. This keeps the report faithful to the project and shows exactly what remains to be demonstrated.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	EcoCycle with a six-item menu	5.1	Fig. 4.3
II	Forms with ≥ 8 distinct field types	Y	Single line, e-mail, number, radio, dropdown, phone, date, multi-line	3.7	Table 3.7
II	Lookup field (relationship between forms)	N	Not seen	4.2	Fig. 4.2
II	Subform (parent–child data)	N	Not seen	4.2	–
II	Field validations / layout rules	Y	Past-date check (reported); mandatory fields	5.6	Table 5.3
II	List report with filter, grouping, custom action	Y	High Priority E-Waste is a filtered list report	5.3	Table 5.1
II	Kanban / Calendar / Timeline / Map report	N	Not seen	5.3	–
II	Pivot table / chart report	Y	Pie chart and two bar charts	5.4	Fig. 5.9–5.11
II	Page / dashboard built with UI Builder	Y	EcoCycle Dashboard	5.4	Fig. 5.9–5.12
II	Deluge – form events (validate / success / field actions)	Y	Validation and successful-submission workflows (reported)	5.6	Table 5.3
II	Deluge – functions, fetch, loops, collections	N	Source not supplied	5.5	Appendix C
II	Email / in-app notifications	Y	High-priority e-mail (reported)	5.6	Table 5.3
II	Form workflow	Y	Four form workflows (reported)	5.6	Table 5.3
II	Scheduled workflow	N	Not seen	5.6	–
II	Approval process	N	Not seen	5.6	–
II	Blueprint	N	Not seen	5.6	–
II	Batch / report workflow	N	Not seen	5.6	–

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Testing & debugging evidence	N	Limited; see Chapter 6	6	Table 6.1
III	Integration with a Zoho service	Y	Zoho CRM	5.7	Fig. 5.13–5.15
III	Connection / REST API call (invokeurl)	Y	Zoho CRM connection	5.7	Fig. 5.13
III	Webhook or Creator REST API	Y	Custom API EcoCycle Summary API (OAuth2); built-in REST API planned	5.17	Fig. 5.18
III	Users, roles and permission profiles	N	Not shown	5.8	–
III	Record-level access criteria / customer portal	N	Not shown	5.8	–
III	Zia / AI feature	Y	EcoCycle Data Assistant; Sentiment field	5.9, 5.16	Fig. 5.16, 5.17
III	Analytics (charts / Zoho Analytics)	Y	Dashboard charts	5.4	Fig. 5.9–5.11
IV	Performance optimisation applied & measured	N	Not measured	7.2	Table 7.2
IV	Deployment across environments	N	Single instance	7.1	Table 7.1
IV	Security controls (encryption, OAuth, permissions)	Y	OAuth2 on the Custom API; read-only chat agent	7.3	Table 7.3
IV	Governance (audit trail, administration)	N	Not described	7.4	–
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used	5.10	–
V	Low-code limitations & spectrum analysis	Y	Sections 8.4 and 8.5	8.4, 8.5	–
—	Mobile app / portal access	N	Not described	5.11	–

Total features implemented: 16 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2		
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6		
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.1, 5.7–5.9		
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7		
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3		

APPENDIX C – COMPLETE DELUGE SOURCE CODE

Listings C.1 to C.4 are reference implementations written from the design described in the project material: the four priority factors, the score ranges, the status defaults, the past-date rule, the high-priority e-mail and the CRM lead. They show how the rules can be coded and are not copies of the scripts in the application. The weights of the priority factors and the field link names (Quantity, Age_years, Contains_Hazardous_Material, Condition, Priority_Score, Priority_Level) are assumed. If the scripts in the application differ, replace these listings with the application's own source.

C.1: Priority calculation and status defaults – DS-01 (reference implementation)

```
// Reference implementation. Form: E-Waste Entry, on successful
submission
    (create).
// The weights below are a reference scheme: they respect the stated
    ranges
// (High 8+, Medium 4-7, Low 0-3) and reproduce the levels seen in the
    test data.
score = 0;
// Quantity
if(input.Quantity >= 8)
{
    score = score + 3;
}
else if(input.Quantity >= 4)
{
    score = score + 2;
}
else if(input.Quantity >= 2)
{
    score = score + 1;
}
// Age in years
if(input.Age_years >= 3)
{
    score = score + 2;
}
else if(input.Age years >= 1)
{
    score = score + 1;
}
// Hazardous material
if(input.Contains_Hazardous_Material == "Yes")
{
    score = score + 3;
}
// Condition
if(input.Condition == "Not Working" || input.Condition == "Damaged")
{
    score = score + 2;
}
if(score >= 8)
{
    level = "High";
}
else if(score >= 4)
```

```
{
    level = "Medium";
}
else
{
    level = "Low";
}
rec = E Waste Entry[ID == input.ID];
rec.Priority_Score = score;
rec.Priority_Level = level;
rec.Collection Status = "Pending";
rec.Recycling_Status = "Not Processed";
```

C.2: Past-date check – DS-02 (reference implementation)

```
// Reference implementation. Form: Collection Request, on validate.
if(input.Preferred_Collection_Date < zoho.currentdate)
{
    alert "The preferred collection date cannot be in the past.";
    cancel submit;
}
```

C.3: High-priority e-mail – DS-03 (reference implementation)

```
// Reference implementation. Form: E-Waste Entry, on successful
    submission,
// after the priority level has been set.
if(input.Priority_Level == "High")
{
    sendmail
    [
        from: zoho.adminuserid
        to: zoho.adminuserid
        subject: "High-priority e-waste: " + input.E_Waste_Name
        message: "Quantity " + input.Quantity + ", collection location
"
            + input.Collection_Location + ", priority score " +
                input.Priority_Score + "."
    ]
}
```

C.4: Create a Zoho CRM lead – DS-04 (reference implementation)

```
// Reference implementation. Form: Collection Request, on successful
    submission.
// The connection name zoho_crm_connection is assumed.
leadMap = Map();
leadMap.put("Last_Name", input.Requester_Name);
leadMap.put("Company", input.Collection_Location);
leadMap.put("Email", input.Email);
leadMap.put("Phone", input.Phone_Number);
resp = zoho.crm.createRecord("Leads", leadMap, Map(),
    "zoho_crm_connection");
info resp;
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open EcoCycle from the Zoho Creator home page and sign in with the Zoho account that has access to it. The menu on the left lists E-waste Entries, App Preferences, Collection Requests, Recycling Tracking, EcoCycle Dashboard and AI Agents.

D.2 Entering E-Waste

- Open E-waste Entries → E-Waste Entry.
- Enter the name, choose the category, enter quantity and age, select whether it contains hazardous material, choose the condition, enter the location and your contact details, and select Submit.
- Open All E-Waste to see the entry; the priority is calculated automatically.

D.3 Requesting Collection

- Open Collection Requests → Collection Request.
- Enter the requester details, location, category, quantity, preferred date and preferred time, and select Submit.
- Use a date that is not in the past. A lead is created in Zoho CRM automatically.

D.4 Tracking Recycling

- Open Recycling Tracking → Recycling Tracking.
- Enter the item name, category, quantity and collection date; update the status, method and recycler name as the work proceeds, and select Submit.
- Open Recycling Tracking Report to review the records.

D.5 Reading the Dashboard and Using the Chat Agent

- Open EcoCycle Dashboard to see the priority pie chart, the collection and recycling status charts and the embedded reports.
- Open AI Agents and type a question, for example: give the e-waste entries with high priority. Review and verify the answer before acting on it.

D.6 Troubleshooting

Table D.1: Troubleshooting Guide

Problem	Likely Cause	Action
Collection request rejected	Preferred date is in the past	Choose today or a later date
No lead in Zoho CRM	Connection not authorised	Check Microservices → Connections and reconnect
Entry not in High Priority E-Waste	Score is below 8	Check quantity, age, hazard and condition
Chat agent returns nothing	No matching records or agent unavailable	Check the report; ask again with different words
Cannot save a form	A mandatory field is blank	Fill every field marked with a red asterisk

D.7 Glossary

Table D.2: Glossary of Terms

Term	Meaning
E-waste	Discarded electrical and electronic equipment such as computers, printers and monitors
Priority score	A number calculated from quantity, age, hazardous material and condition
Priority level	High (8 or above), Medium (4 to 7) or Low (0 to 3)
Lead	A record of a possible customer or contact in Zoho CRM
Connection	A stored authorisation that lets Zoho Creator call another service
Chat agent	A Zia agent that answers questions in a chat page
Deluge	The scripting language of Zoho Creator

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	JENELL S G	Register No.	240701212
Project Title	ECOCYCLE – E-WASTE COLLECTION & RECYCLING MANAGEMENT	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner	**Signature of External Examiner**
Name: _____	Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	**JENELL S G**	**Register No.**	**240701212**
Department / Section	CSE	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	**Question / Topic Asked**	**Unit**	**Response (E / G / S / NI)**	**Remarks**
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	**Max. Marks**	**Marks Awarded**
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER	**EXTERNAL EXAMINER**
Name: _____	Name: _____